

Lem transformada fourier conmutativa integral

Lema 1. Sean $f, g \in \mathcal{L}^1(\mathbb{R})$, entonces $\int_{\mathbb{R}} f(y) g^\wedge(y) \, dy = \int_{\mathbb{R}} f^\wedge(x) g(x) \, dx$.

Demostración: Como $g^\wedge \in \mathcal{L}^\infty(\mathbb{R})$ (ver obs-propiedades-transformada-fourier-11/Observación 1), el producto $f \cdot g^\wedge \in \mathcal{L}^1(\mathbb{R})$. De manera análoga, $f^\wedge \cdot g \in \mathcal{L}^1(\mathbb{R})$. Entonces, por el Teorema de Fubini, se tiene que

$$\begin{aligned} \int_{\mathbb{R}} f(y) g^\wedge(y) \, dy &= \int_{\mathbb{R}} f(y) \left(\int_{\mathbb{R}} g(x) e^{-2\pi i x y} \, dx \right) \, dy \\ &\stackrel{\text{Fubini}}{=} \int_{\mathbb{R}} g(x) \left(\int_{\mathbb{R}} f(y) e^{-2\pi i x y} \, dy \right) \, dx = \int_{\mathbb{R}} f^\wedge(x) g(x) \, dx. \end{aligned}$$

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